

Chengxiang Huang

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EDUCATION BACKGROUND

Beijing University of Posts and Telecommunications & Queen Mary University of London

B.Eng. in Telecommunication Engineering with Management

Sept. 2021 – June 2025

GPA: 90.35/100 (BUPT); 76/100 First Class Honours (QMUL)

Rank: 9 / 320

TOEFL: 108

Course Highlights: Principles of Communication: 97; Advanced Java Programming: 95; Electromagnetic Fields and Waves: 93; Machine Learning: 92; Internet Protocols and Networks: 92; Programming Fundamentals: 90...

Awards and Activities:

1. Outstanding Camper, VEX Robotics Winter Camp, Shanghai Jiao Tong University
2. First-Class University-Level Scholarship

PUBLICATIONS

- [1] **Chengxiang Huang**, Yake Wei, Zequn Yang and Di Hu, “Adaptive Unimodal Regulation for Balanced Multimodal Information Acquisition,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. [\[arXiv\]](#) [\[code\]](#)
- [2] **Chengxiang Huang**, Bingyan Liu, “PA³Fed: Period-Aware Adaptive Aggregation for Improved Federated Learning,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025.
- [3] Shaoxuan Xu, **Chengxiang Huang**, Menglu Cui, Hongfa Wang, Di Hu, “BalanceBenchmark: A Survey for Multimodal Imbalance Learning,” submitted to *NeurIPS2025*, D&B track. [\[arXiv\]](#) [\[code\]](#)

RESEARCH EXPERIENCE

Renmin University of China - Gaoling School of Artificial Intelligence - GeWu Lab — Intern

Beijing | April 2024 to present

- **Research Areas:** Conducting in-depth studies in artificial intelligence, focusing on multimodal learning. Leveraging information theory to explore the mechanism of information acquisition across different modalities in multimodal learning.
- **Mechanism Exploration:** Investigated how varying modalities contribute differently to the learning process, identifying key factors that influence information sufficiency and balance in multimodal models.

Louisiana State University (LSU) - Department of Computer Science - Pisces Lab — Intern

Remote | January 2024 to May 2024

- **Research Focus:** Conducted an in-depth study on federated learning, focusing on addressing the challenges posed by non-iid (non-independent and identically distributed) data across clients and its impact on federated learning performance.
- **Skill Development:** Gained proficiency in reading and analyzing academic papers, developing foundational skills in deep learning and machine learning.

PROJECT

Maze Treasure Hunting Car Development — Team Leader / Algorithm Developer BUPT, July 2023

- **Tech Stack:** Proficiently applied PID control algorithm for path planning and speed adjustment, combined with YOLO object detection technology for real-time obstacle and target recognition within the maze.
- **System Integration:** Integrated control algorithms with sensor data, developed the car's autonomous navigation system, ensuring efficient navigation in complex maze environments.
- **Team Collaboration:** Collaborated closely with team members, achieving a score of 96/100 in the maze treasure-hunting challenge.